is a filler cap fitted for pouring the brake oil and it is closed tightly. In the filler cap a small air vent hole is provided for maintaining the atmospheric air pressure in the reservoir. The sectional view of the master cylinder can be seen in Figure 6.4.1.3. The piston travels front and back inside the reservoir chamber. The push rod and brake pedal are connected with each other.

As the return or helical spring is placed in front of the piston, the piston always stays in the direction where the push rod stays. With this arrangement a check valve is also connected with four wheel adapter for transferring brake fluid to four wheels. In addition to this operating system a brake light switch is also connected. There are two ports called as inlet port and bypass port located on the wall which separates reservoir tank and compressor unit. The inlet port is made as slightly larger, whereas the bypass port is smaller in their construction. In the front face of the piston small feed holes are provided. In front of the feed holes a primary rubber cup is also present.

These feed holes allow the brake fluid to move from backside of the piston to the front. A check valve and a return spring are provided in front of the primary rubber cup. A secondary rubber cup is also located at the back side of the piston for avoiding brake fluid leakages. In this arrangement, the check valve and primary rubber cup arrangement act as one way valve.

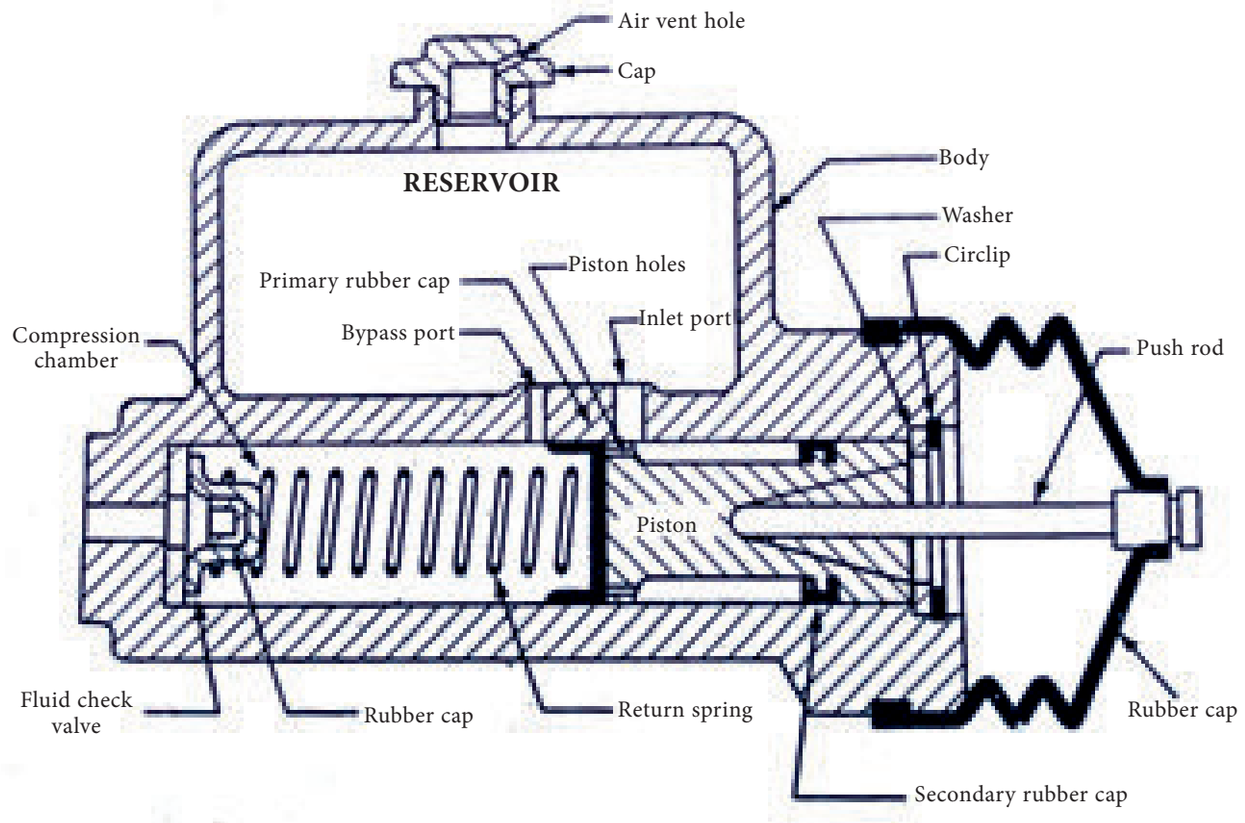


Figure 6.4.1.2 (a) Sectional view of a master cylinder

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